

Fig. 6. Discharge–depth scattergrams for miniature flumes at S1 and S2, least-squares regression equations fitted to these data and discharge–depth rating equations used in the determination of discharges through S1 and S2.

possessing a range of surface roughnesses. As a consequence, flows of comparable depth at different flumes may have quite dissimilar discharges. Rating equations were determined from these data by performing ordinary least-squares regressions and then correcting the regression equations by the method described by Ferguson (1986). This method eliminates the bias arising

TABLE 1

Discharge through S1 and S2 at 5 min intervals during the four experiments

Experiment and minute	Discharge at S1 ($cm^3\ s^{-1}$)	Discharge at S2 ($cm^3\ s^{-1}$)	Revised discharge at S2 ($cm^3\ s^{-1}$)
E1-4	1139	779	505
E1-9	3131	2652	2632
E2-4	1594	1482	1201
E2-9	2871	3366	3631
E2-14	3397	3851	4354
E2-19	3451	4160	4979
E3-4	2333	3143	3311
E3-9	3212	4395	5203
E4-4	2742	2301	2174
E4-9	3028	3317	3559
E4-14	3398	4289	5035
E4-19	3585	4674	5654
E4-24	3371	4442	5278
E4-29	3746	4453	5296